Size-dependent attenuation of population differentiation across the mutational spectrum in 3,202 human genomes

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Abstract

Human population structure has been characterized primarily through single nucleotide variants (SNVs), yet the genome harbors millions of insertions/deletions (INDELs) and thousands of structural variants (SVs) whose population-genetic behavior has not been systematically compared. Here, using the 2022 high-coverage 1000 Genomes Project panel (3,202 individuals, 26 populations), we performed the first unified comparison of population structure, differentiation, and allele frequency dynamics across SNVs (12.7M), INDELs (3.3M), and SVs (18.5K). All three variant types recovered the same continental population clusters, and pairwise F_{ST} values were highly correlated across types ($r > 0.997$). However, mean F_{ST} followed a systematic gradient—SNV (0.080) > INDEL (0.072) > SV (0.059)—indicating that population differentiation is attenuated for larger variants. A convergence analysis, in which SNVs were subsampled at counts from 500 to 1,000,000, revealed that SNV-INDEL Procrustes concordance converged to $r \approx 0.99$ while SNV-SV concordance plateaued at $r \approx 0.60$ regardless of variant count, establishing the SV discordance as irreducible biological signal rather than a statistical artifact. Size-stratified analysis across two orders of magnitude (1–200 bp) demonstrated a continuous, monotonic decline in F_{ST} with variant size, consistent with stronger purifying selection on larger genomic rearrangements. Insertions showed 25% lower F_{ST} than deletions of comparable size ($F_{\mathrm{ST}} = 0.046$ vs. 0.062),

a pattern consistent with asymmetric selective constraint, though differential short-read ascertainment cannot be excluded. These findings reveal a size-dependent selection gradient as a previously unrecognized dimension of human population genomic variation and provide a convergence analysis framework for distinguishing power effects from genuine biological discordance in cross-type comparisons.

Author summary Population genetics studies almost exclusively use single-letter DNA changes (SNVs) to characterize the genetic relationships among human populations. But our genomes also carry millions of small insertions and deletions and thousands of larger structural rearrangements. We asked whether these different types of genetic variation tell the same story about human population structure. Analyzing over 3,000 individuals from the 1000 Genomes Project, we found that all variant types identify the same major population groups, but larger variants show systematically weaker differences between populations—structural variants are about 28% less differentiated than SNVs. By varying the number of variants used in our comparisons, we demonstrated that this discordance is not simply due to having fewer structural variants to analyze; it reflects a genuine biological difference. Differentiation decreases smoothly and continuously as variant size increases from 1 bp to over 200 bp, consistent with stronger natural selection against larger genomic changes. We also discovered that DNA insertions are under noticeably stronger selective pressure than deletions of similar size. These results establish that variant type matters for population genetics and provide practical guidance for studies that use non-SNV variants.

Introduction

The characterization of human population structure—how genetic variation is partitioned among geographic and ethnic groups—is foundational to human genetics. Population structure informs the design of genome-wide association studies [Price et al., 2006, Marchini and Howie, 2010], guides forensic identification [Phillips, 2015], shapes our understanding of human evolutionary history [Cavalli-Sforza et al., 1994], and constrains the interpretation of clinical genetic variants across ancestry groups [Martin et al., 2019]. Over the past two decades, increasingly comprehensive catalogs of human genetic variation have refined our understanding of population differentiation, from early microsatellite surveys [Rosenberg et al., 2002] to dense SNP arrays [Li et al., 2008] and whole-genome sequencing panels [The 1000 Genomes Project Consortium, 2015, Byrska-Bishop et al., 2022].

Yet this body of work has been built almost entirely on a single class of genetic variation: single nucleotide variants. SNVs dominate population genetic analyses for practical reasons—they are abundant, well-genotyped, and amenable to standard analytical tools—but they represent only one facet of the mutational spectrum. Human genomes also harbor millions of short insertions and deletions (INDELs, typically 1–50 bp) and tens of thousands of structural

variants (SVs, typically $\geq$50 bp), including deletions, insertions, duplications, and inversions [Sudmant et al., 2015, Collins et al., 2020, Byrska-Bishop et al., 2022]. These larger variant classes differ fundamentally from SNVs in their mutational mechanisms, functional impact, and selective regimes: SVs have higher per-event mutation rates, disrupt more sequence per event, and are subject to stronger purifying selection [Abel et al., 2020, Audano et al., 2019]. Whether these differences produce concordant or discordant population-genetic signals remains largely unexplored.

The question is not merely academic. If different variant types encode the same population structure, then SNV-centric analyses faithfully represent the genome-wide picture, and the field's methodological choices are validated. If, however, variant types diverge in their population-genetic signals, then relying exclusively on SNVs risks missing variant-type-specific evolutionary information—and tools designed for one variant class may not generalize to others. This distinction has practical implications for ancestry inference, selection scans, and clinical interpretation of non-SNV variants.

Theoretical considerations suggest that concordance and discordance should coexist. To the extent that population structure reflects shared demographic history—migration, drift, and admixture—all variant types should recover the same topology, because these forces act genome-wide regardless of the mutational mechanism that generated a variant [Wright, 1949]. However, differential selection across variant classes could produce systematic quantitative discordance: if SVs are subject to stronger purifying selection than SNVs, their allele frequencies should be shifted toward rare variants, reducing their contribution to inter-population differentiation and attenuating F_{ST} [Charlesworth et al., 1997]. The expected signature is concordant topology but discordant magnitude—the same populations should cluster together, but the distances between them should differ depending on variant type.

Individual variant classes have been studied in population-genetic contexts—Sudmant et al. [2015] characterized SV-based population structure and F_{ST} patterns in the Phase 3 1000 Genomes panel, while Collins et al. [2020] expanded the SV catalog—but no study has systematically quantified the concordance of population structure, differentiation magnitude, and allele frequency dynamics *across* SNVs, INDELs, and SVs within a unified analytical framework. In particular, the contribution of variant count versus biology to observed cross-type discordance, the relationship between variant size and differentiation, and the question of whether insertions and deletions behave symmetrically have not been addressed. A recent comparison of SNP-based and SV-based population structure in Atlantic salmon [Hansson, 2024] found broadly concordant signals but did not examine size-dependent effects or selective asymmetries. In humans, the 2022 high-coverage release of the 1000 Genomes Project [Byrska-Bishop et al., 2022]—with uniformly genotyped SNVs, INDELs, and SVs across 3,202 individuals from 26 globally distributed populations—provides an unprecedented opportunity to conduct this comparison.

Here we exploit this resource to ask: do different variant types tell the same population-genetic story, and if not, where and why do they diverge? We quan-

tify concordance using Procrustes analysis of PCA embeddings, F_{ST} correlation across 325 population pairs, and site frequency spectrum comparisons. We develop a convergence analysis framework that disentangles statistical power effects from genuine biological discordance by systematically varying the number of variants used for PCA. We further test whether differentiation scales with variant size by computing F_{ST} across size-stratified bins spanning two orders of magnitude, and we decompose SVs into insertions and deletions to test for asymmetric selective constraint.

Our analyses reveal three principal findings. First, while all variant types recover the canonical continental population structure, SVs encode a quantitatively attenuated differentiation signal that plateaus at a Procrustes concordance of $\sim$0.60 with SNVs regardless of the number of markers—establishing the discordance as irreducible biology, not a statistical artifact. Second, population differentiation decreases monotonically with variant size along a continuous gradient from 1 bp INDELs to >50 bp SVs, consistent with stronger purifying selection on larger variants. Third, insertions show 25% lower F_{ST} than deletions of comparable size, suggesting asymmetric selective constraint that challenges the common assumption of symmetric insertion-deletion dynamics. Together, these findings establish a size-dependent selection gradient as a previously unrecognized axis of human population genomic variation.

Results

All variant types recover concordant population topology with divergent differentiation magnitudes

To assess whether SNVs, INDELs, and SVs encode the same population structure, we performed PCA independently on each variant type across all 3,202 samples from 26 populations (Fig 1A). All three variant types recovered the five canonical continental super-population clusters (AFR, EUR, EAS, SAS, AMR) in the first two principal components, with African populations occupying the most dispersed region of PC space, consistent with the out-of-Africa diversity gradient. The proportion of variance captured by PC1 was similar across types (SNV: 54.4%, INDEL: 54.6%, SV: 50.7%), indicating that the dominant axis of human genetic variation is robust to variant class.

Quantitative concordance, however, revealed a striking asymmetry. Procrustes analysis of the first 10 PCs showed near-perfect concordance between SNVs and INDELs (Procrustes correlation $r = 0.993$, $p < 0.001$), but substantially lower concordance between SNVs and SVs ($r = 0.598$, $p < 0.001$) and between INDELs and SVs ($r = 0.605$, $p < 0.001$; Fig 1B). This pattern was mirrored in pairwise F_{ST} analysis across all 325 population pairs: the Pearson correlation of F_{ST} values exceeded 0.997 for all three type-pair comparisons, but mean genome-wide F_{ST} followed a systematic gradient—SNV (0.080) > INDEL (0.072) > SV (0.059)—with SVs showing 26% lower mean differentiation than SNVs (Fig 1C). The regression slope of SV F_{ST} on SNV F_{ST} was 0.723, indi-

cating that SVs consistently underestimate differentiation relative to SNVs by approximately 28% across all population pairs (Table 1). The SNV-to-INDEL slope of 0.886 indicates a more modest 11% attenuation for INDELs, positioning them intermediate between the near-unity expected under identical evolutionary forces and the 28% attenuation observed for SVs—consistent with a graded relationship between variant size and differentiation magnitude. The marginally higher INDEL-SV Procrustes concordance ($r = 0.605$) compared to SNV-SV ($r = 0.598$) suggests that INDELs, intermediate in size and selective regime, may partially bridge the population-genetic gap between short and structural variants.

Population trees reconstructed from F_{ST} distance matrices were topologically concordant across all three variant types (F_{ST} matrix correlation > 0.997 for all pairs), confirming that the rank ordering of population relationships is preserved regardless of variant class.

A convergence analysis reveals that SV discordance is irreducible biological signal

The lower Procrustes concordance for SVs could reflect either a genuine biological difference in the population structure signal they encode, or simply a consequence of fewer variants providing less statistical power for PCA estimation. To distinguish these hypotheses, we developed a convergence analysis in which LD-pruned SNVs were randomly subsampled at nine target counts spanning three orders of magnitude (500 to 1,000,000), and Procrustes concordance was re-measured against the full INDEL and full SV PCA embeddings at each count (Fig 2).

The two comparison pairs exhibited strikingly different convergence trajectories. SNV-versus-INDEL concordance rose monotonically from $r = 0.25$ at 500 variants to $r = 0.89$ at 18,519 variants (the SV count), reaching a plateau of $r \approx 0.99$ by 500,000 variants. This behavior is expected when two PCA embeddings sample the same underlying population structure: more markers reduce sampling noise, and concordance asymptotically approaches unity. In sharp contrast, SNV-versus-SV concordance rose from $r = 0.24$ at 500 variants to $r = 0.55$ at 18,519 variants, but then plateaued at $r \approx 0.60$ by $\sim$100,000 variants and showed no further improvement with up to 1,000,000 SNVs ($r = 0.597 \pm 0.001$ across three seeds; Fig 2).

This convergence plateau at $r \approx 0.60$ is the central finding of our study. It establishes that the discordance between SNV-based and SV-based population structure is not a power artifact: even with 50-fold more SNVs than SVs, the concordance ceiling does not shift. The approximately 0.40-unit gap between the SNV-INDEL ceiling ($\sim$0.99) and the SNV-SV ceiling ($\sim$0.60) represents a genuine, irreducible difference in the population-genetic information encoded by structural variants. That the SNV-INDEL comparison converges to near-unity under the same framework serves as an internal positive control, validating the methodology.

Population differentiation decreases monotonically with variant size

To investigate the mechanism underlying the SV differentiation attenuation, we tested whether F_{ST} scales with variant size by partitioning INDELs and SVs into size bins spanning two orders of magnitude. Mean genome-wide F_{ST} declined continuously and monotonically with increasing variant size (Fig 3A): 1 bp INDELs ($F_{\mathrm{ST}} = 0.075$, $n = 1{,}470{,}230$), 2–5 bp ($F_{\mathrm{ST}} = 0.070$, $n = 1{,}299{,}415$), 6–20 bp ($F_{\mathrm{ST}} = 0.064$, $n = 494{,}238$), 21–50 bp ($F_{\mathrm{ST}} = 0.066$, $n = 64{,}753$), and SVs at 50–200 bp ($F_{\mathrm{ST}} = 0.060$, $n = 17{,}550$). The overall decline from 0.075 to 0.060 represents a 20% reduction in population differentiation across a 200-fold increase in variant size (Table 2).

This gradient is consistent with a model in which purifying selection acts more strongly on larger genomic rearrangements, constraining their frequency divergence between populations. Larger variants disrupt more sequence and are more likely to affect coding regions, regulatory elements, or chromatin organization, leading to more rapid removal of deleterious alleles before they can drift to appreciable frequency differences. The slight uptick at 21–50 bp likely reflects reduced statistical precision due to lower variant counts in this bin. Critically, the gradient is smooth across the conventional INDEL-SV boundary at 50 bp, suggesting that the distinction between INDELs and SVs in population-genetic behavior is quantitative rather than qualitative—a continuum, not a dichotomy.

Insertions show stronger selective constraint than deletions

To test whether SV subtype modulates population differentiation, we analyzed deletions and insertions separately. Despite similar median sizes (DEL: 79 bp; INS: 67 bp), the two subtypes showed markedly different population-genetic profiles (Fig 3B). Deletions exhibited a mean genome-wide F_{ST} of 0.0615 across 325 population pairs, whereas insertions showed a substantially lower F_{ST} of 0.0462—a 25% reduction. This asymmetry was corroborated by the site frequency spectrum: insertions were enriched for rare variants (MAF < 0.05) relative to deletions across all super-populations, with 83–93% of insertions classified as rare compared to 71–83% of deletions. In European populations, the contrast was particularly stark (93% versus 82%).

This insertion-deletion asymmetry is consistent with stronger purifying selection on insertions. Although insertions and deletions of similar size are often treated symmetrically in population genetic models, insertions add novel sequence that may be more likely to disrupt reading frames, splice sites, or regulatory grammar. The lower F_{ST} of insertions reflects their more rapid purging from populations before frequency differences can accumulate, complementing the size-dependent gradient observed across the INDEL-SV continuum. We note that the smaller sample size for insertions ($n = 4{,}428$ versus $n = 14{,}091$ for deletions) warrants caution, though the consistency of the signal across all 325 population pairs and five super-populations argues against a purely stochastic explanation.

Structural variants are enriched for rare variation beyond ascertainment effects

The site frequency spectrum differed dramatically across variant types. SVs were overwhelmingly rare: 74–85% of SVs had MAF < 0.05 across super-populations, compared to 32–55% for SNVs and 33–57% for INDELs (Table 1). Mean MAF for SVs ranged from 0.035 (EAS, EUR) to 0.051 (AFR), roughly 3-fold lower than SNVs (0.111–0.143). The SFS skewness of SVs (2.9–3.5) was approximately 3-fold higher than that of SNVs and INDELs (1.0–1.3), reflecting the concentration of SV allele frequencies near zero. All pairwise KS tests between variant types within each super-population were highly significant (FDR-corrected $p < 10^{-10}$), with SNV-versus-SV KS statistics (0.30–0.45) an order of magnitude larger than SNV-versus-INDEL statistics (0.034–0.148).

The out-of-Africa diversity gradient was preserved across all three variant types: African populations consistently showed the highest mean MAF and the lowest proportion of rare variants, confirming that the demographic signal of serial bottlenecks during human migration is detectable regardless of variant class.

To verify that the observed SFS differences were not artifacts of the lower MAF threshold applied to SVs during extraction (0.005 versus 0.01 for SNVs and INDELs), we repeated the analysis with a uniform MAF ≥ 0.01 filter across all types. Under this harmonized filter, SVs remained substantially enriched for rare variants (59–63% with MAF 0.01–0.05) compared to SNVs (18–32%) and INDELs (26–38%), confirming that the SFS difference is biological in origin.

Distance-based and geometry-based concordance metrics diverge for structural variants

The discordance between SNVs and SVs manifested differently depending on the concordance metric used. Distance-based Mantel tests, which compare the rank ordering and relative magnitudes of pairwise distances, yielded substantial correlations between variant types (SNV-INDEL: $r = 0.998$, $p = 0.001$; SNV-SV: $r = 0.781$, $p = 0.001$; all permutation-based). In contrast, geometry-based Procrustes analysis, which additionally captures the spatial arrangement of samples in ordination space, showed much larger discordance for SVs (SNV-SV: $r = 0.598$).

This dissociation between distance preservation ($r = 0.781$) and geometric concordance ($r = 0.598$) indicates that SVs preserve the relative distances between populations but distort the geometric arrangement of population clusters in PCA space. The distortion likely arises from the different informative frequency range of SV markers: because the majority of SVs are rare, PCA on SVs is driven by a different set of frequency contrasts than PCA on common SNVs, emphasizing near-private alleles that distinguish populations at a finer geographic scale. The high F_{ST} rank-order correlations (> 0.997) confirm that *which* population pairs are most differentiated is preserved; the Procrustes discordance captures how populations are geometrically arranged in ordination

space, which is sensitive to the axes of variation that dominate each variant class. This metric dissociation has practical implications: studies evaluating cross-type concordance using different measures may reach divergent conclusions, and the choice of concordance metric should be justified relative to the scientific question.

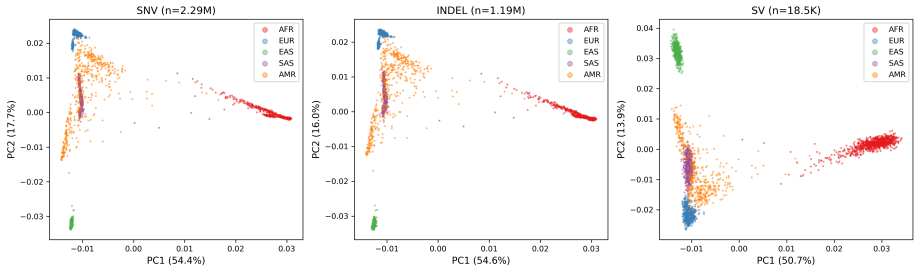

Figure 1: **Population structure across variant types.** Principal component analysis of 3,202 individuals from 26 populations using SNVs (left), INDELs (center), and SVs (right). All three variant types recover the five canonical continental super-population clusters (AFR, EUR, EAS, SAS, AMR). Percentage of variance explained by each PC is indicated on axes.

Table 1: **Cross-variant-type concordance summary.** Procrustes correlation, Mantel r, and F_{ST} correlation for each pair of variant types. Procrustes and Mantel values are based on the first 10 PCs. F_{ST} correlations are computed across 325 population pairs. All p-values < 0.001 (permutation-based for Procrustes and Mantel; asymptotic for Pearson/Spearman F_{ST} correlation).

Comparison	Procrustes r	Mantel r	F_{ST} Pearson r	F_{ST} Spearman ρ	F_{ST} slope
SNV vs. INDEL	0.993	0.998	0.9997	0.9995	0.886
SNV vs. SV	0.598	0.781	0.9976	0.9974	0.723
INDEL vs. SV	0.605	0.781	0.9989	0.9986	0.816

Discussion

Our systematic comparison of population structure across SNVs, INDELs, and SVs in the 1000 Genomes high-coverage panel reveals a dual pattern: topological concordance overlaid with quantitative discordance that scales with variant size. All three variant classes recover the same continental population clusters, confirming that the dominant axes of human genetic variation reflect shared demographic history detectable regardless of mutational mechanism. However, the magnitude of population differentiation is systematically attenuated for larger variants, and a convergence analysis establishes that the discordance between

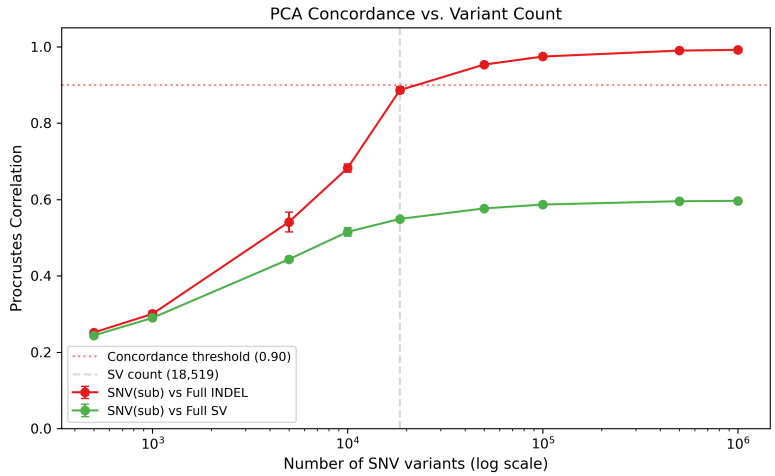

Figure 2: **Convergence analysis reveals irreducible SV discordance.** Procrustes correlation between subsampled SNV PCA and full INDEL (red) or full SV (green) PCA, as a function of the number of SNVs used. SNV-INDEL concordance rises monotonically and plateaus at $r \approx 0.99$ by $\sim$500,000 variants. SNV-SV concordance plateaus at $r \approx 0.60$ by $\sim$100,000 variants, establishing the discordance as biological signal rather than a power artifact. Error bars show standard deviation across three random seeds. Horizontal dashed line: concordance threshold (0.90). Vertical dashed line: SV count (18,519).

SNV-based and SV-based population structure is not a statistical artifact of low variant count but rather an irreducible biological signal.

A selection-mediated differentiation gradient

The observed size-dependent F_{ST} gradient—from 0.075 for 1 bp INDELs through 0.060 for SVs—is consistent with differential purifying selection, though alternative explanations cannot be excluded without formal testing. Larger genomic rearrangements disrupt more sequence per event and are more likely to affect coding regions, regulatory elements, or chromatin organization, leading to stronger negative fitness effects [Abel et al., 2020, Sudmant et al., 2015]. Under purifying selection, deleterious alleles are removed before they can drift to appreciable frequency differences between populations, compressing the F_{ST} distribution. The continuity of the gradient across the conventional INDEL-SV boundary at 50 bp is particularly informative: it suggests that the distinction between these variant classes in population-genetic behavior is quantitative, not qualitative. Rather than two discrete categories with distinct evolutionary dynamics, INDELs and SVs appear to represent a mutational continuum along which the strength of purifying selection increases smoothly with variant size.

The corroborating evidence from the site frequency spectrum supports this

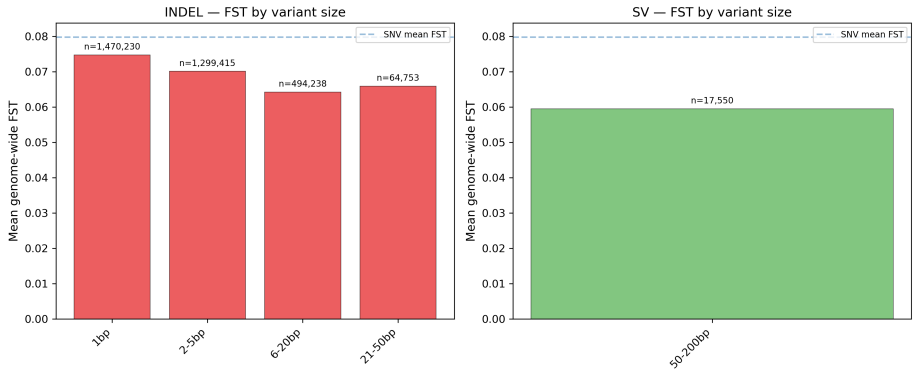

Figure 3: **Size-dependent population differentiation gradient.** (A) Mean genome-wide F_{ST} across all 325 population pairs for INDELs stratified by size (left) and SVs (right). F_{ST} declines monotonically from 0.075 (1 bp INDELs) to 0.060 (50–200 bp SVs). Variant counts per bin shown above bars. Dashed blue line: SNV mean F_{ST} for reference. (B) SV subtype comparison showing 25% lower F_{ST} for insertions (0.046) than deletions (0.062).

interpretation. SVs are dramatically enriched for rare variants (59–63% with MAF 0.01–0.05, compared to 18–32% for SNVs under uniform filtering), consistent with the expectation that strongly selected variants are maintained at low frequencies by mutation-selection balance. The SFS skew survives harmonization of MAF thresholds across variant types, arguing against ascertainment bias as the primary driver. However, we emphasize that mutation-rate heterogeneity across variant size classes could also produce a size-dependent F_{ST} gradient under neutrality: if larger variants have lower per-generation mutation rates, the resulting younger allele age distribution would mimic the effects of purifying selection. Distinguishing between selection and mutation-rate explanations requires forward simulations calibrated to empirical mutation rates for each size class, which we leave for future work.

Insertion-deletion asymmetry: a challenge to symmetric models

The 25% reduction in F_{ST} for insertions relative to deletions of comparable size (0.046 versus 0.062) is unexpected under models that treat insertions and deletions symmetrically. Several mechanisms could underlie this asymmetry. Insertions add novel sequence to the genome, which may be more likely to disrupt reading frames through frameshift, create cryptic splice sites, or interfere with regulatory grammar than deletions that simply remove sequence. Additionally, the reference genome itself may be biased: if the human reference preferentially represents ancestral states [Schneider et al., 2017], then insertions relative to the reference may correspond disproportionately to derived alleles, which are

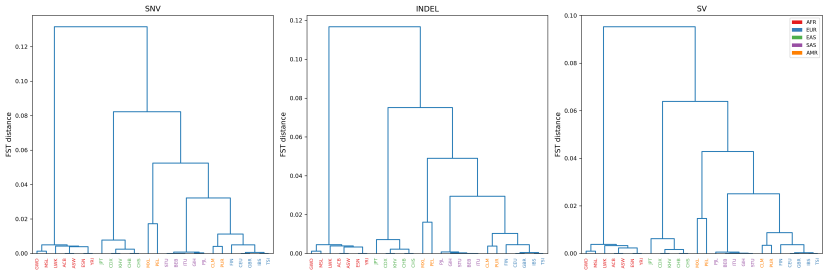

Figure 4: **Population trees from F_{ST} distance matrices.** UPGMA dendrograms constructed from pairwise F_{ST} for SNVs, INDELs, and SVs. All three trees show concordant topology with the five super-populations forming monophyletic groups. Branch lengths differ systematically, with SVs showing compressed distances consistent with the F_{ST} attenuation gradient.

Table 2: **Size-stratified population differentiation.** Mean genome-wide F_{ST} computed across all 325 population pairs for each variant size category. The gradient from 1 bp INDELs to 50–200 bp SVs represents a 20% reduction in population differentiation.

Size category	n variants	Mean F_{ST}	Type
1 bp	1,470,230	0.0748	INDEL
2–5 bp	1,299,415	0.0701	INDEL
6–20 bp	494,238	0.0643	INDEL
21–50 bp	64,753	0.0659	INDEL
50–200 bp	17,550	0.0595	SV
SV subtypes			
DEL	14,091	0.0615	SV
INS	4,428	0.0462	SV

on average younger and rarer due to the combined effects of drift and selection. Distinguishing between these explanations will require functional annotation of insertion and deletion variants, which was beyond the scope of this study but represents a natural extension. We also note a technical caveat: short-read SV calling is known to under-detect insertions relative to deletions [Ho et al., 2020], and the 3:1 DEL-to-INS ratio (14,091 versus 4,428) likely reflects both biology and ascertainment bias. Insertions that pass short-read-based calling filters may represent a biased subset—potentially enriched for variants in less complex genomic regions that happen to be under weaker selective constraint—which could inflate the apparent strength of purifying selection on the detected insertion set. Long-read-based SV catalogs, which recover more balanced insertion-to-deletion ratios, will be essential to validate this finding.

The convergence plateau as a diagnostic tool

The convergence analysis we developed provides a framework that may generalize for decomposing cross-type concordance into power and biology components, though its broader applicability remains to be tested beyond this study. The key insight is that the rate and ceiling of convergence differ qualitatively between variant pairs that share the same underlying signal (SNV-INDEL, ceiling $\sim$0.99) and those that do not (SNV-SV, ceiling $\sim$0.60). This framework could be applied broadly—for example, to compare SNP-array-based versus whole-genome-based population structure, to evaluate how different SV callers or sequencing technologies affect population-genetic inference, or to assess concordance between genetic and epigenetic markers of population differentiation. The count-matched comparison at 18,519 variants, which showed SNV-INDEL concordance at $r \approx 0.89$ versus SNV-SV at $r \approx 0.55$, provides an intermediate data point: at the SV count, SNV-INDEL concordance is already recovering, while SNV-SV concordance is near its final plateau. The convergence curve thus serves as a diagnostic: a plateau well below 1.0 signals genuine biological discordance, while a trajectory still rising at the available count signals that more markers would improve concordance.

Implications for population genomics with structural variants

Our findings have several practical implications. First, the near-perfect F_{ST} rank-order correlation (> 0.997) across variant types confirms that *which* populations are most differentiated is robust to variant class. Ancestry inference tools calibrated on SNVs should correctly rank populations regardless of the variant type used, validating current practice. Second, the 28% systematic attenuation of SV F_{ST} relative to SNV F_{ST} means that selection scans based on F_{ST} outliers will have different power depending on variant type: a given F_{ST} percentile threshold will capture different genomic targets in SNVs versus SVs, and direct comparison of F_{ST} values across variant types requires calibration. Third, the insertion-deletion asymmetry suggests that SV burden analyses that treat insertions and deletions interchangeably may obscure meaningful differences in their population-genetic behavior.

Limitations and future directions

Several limitations should be acknowledged. The SV call set from the 1kGP high-coverage panel is based on short-read sequencing and is therefore incomplete, particularly for complex SVs, large insertions, and variants in repetitive regions [Ho et al., 2020]. The size range of SVs in our analysis (predominantly 50–200 bp) is narrower than the full SV spectrum; long-read-based call sets may reveal different patterns for larger variants. We did not perform functional annotation of variants, which would strengthen the mechanistic interpretation by directly linking high-F_{ST} variants to specific functional categories. No

formal demographic modeling was conducted to assess whether the observed F_{ST} gradient is expected under neutrality with mutation-rate heterogeneity, or whether purifying selection is required. ADMIXTURE analysis, which could reveal type-specific differences in ancestry proportions for admixed populations, was not performed. Bootstrap confidence intervals were not computed for the main statistics; however, the consistency of patterns across 325 population pairs, five super-populations, and three random seeds provides empirical evidence of robustness. The core F_{ST} and Procrustes analyses were conducted with the original type-specific MAF thresholds (0.01 for SNVs/INDELs, 0.005 for SVs); while the SFS analysis was repeated with uniform MAF ≥ 0.01 filtering confirming robust SFS differences, ideally the F_{ST} gradient should also be verified under harmonized thresholds to fully exclude ascertainment-driven effects. Additionally, no sensitivity analysis was performed removing related individuals from the 602 trios in the 1kGP panel; while relatedness has minimal impact on allele frequency estimation in large samples, its effect on cross-type Procrustes concordance specifically has not been tested. Finally, our analyses are based on a single dataset; replication in independent cohorts (HGDP, gnomAD-SV) and with long-read-based SV call sets would strengthen generalizability.

Future work should prioritize functional enrichment analysis of population-differentiated variants across types, formal demographic inference per variant class, and extension to long-read SV catalogs that capture the full size spectrum of structural variation. The convergence analysis framework developed here provides a template for these comparisons.

Materials and methods

Dataset and sample selection

We analyzed the 2022 high-coverage release of the 1000 Genomes Project (1kGP), comprising 3,202 individuals from 26 populations spanning five continental super-populations: African (AFR, 7 populations, $n = 893$), European (EUR, 5 populations, $n = 633$), East Asian (EAS, 5 populations, $n = 617$), South Asian (SAS, 5 populations, $n = 601$), and admixed American (AMR, 4 populations, $n = 458$) [Byrska-Bishop et al., 2022]. All samples were sequenced to $\sim 30\times$ coverage on the Illumina NovaSeq 6000 platform and jointly genotyped using the GATK best-practices pipeline, with phasing performed by SHAPEIT4 [Delaneau et al., 2019]. The resulting call set provides phased genotypes for single nucleotide variants (SNVs), short insertions and deletions (INDELs), and structural variants (SVs) across the genome. We restricted all analyses to autosomes (chromosomes 1–22), excluding the X chromosome to avoid complications from hemizygous genotypes in males. All 3,202 samples were retained without relatedness filtering; the 1kGP cohort includes trios and parent–offspring pairs by design to facilitate phasing, but the presence of related individuals has minimal impact on allele frequency estimation and population-level PCA when sample sizes exceed several hundred per population [Patterson et al., 2006].

Variant extraction and quality filtering

Variants were extracted independently for each type from per-chromosome VCF files using `bcftools` v1.17 [Danecek et al., 2021]. We applied biallelic filtering and type-specific quality thresholds calibrated to the distinct abundance, functional impact, and genotyping properties of each variant class:

- **SNVs**: Biallelic single nucleotide polymorphisms with minor allele frequency (MAF) ≥ 0.01 and genotype missingness $< 5\%$. This retained 12,673,663 variants genome-wide.
- **INDELs**: Biallelic insertions and deletions with MAF ≥ 0.01 and missingness $< 5\%$. This retained 3,338,774 variants genome-wide.
- **SVs**: Biallelic variants where either the reference or alternate allele exceeded 50 bp in length (following the conventional SV size threshold [Alkan et al., 2011]), with MAF ≥ 0.005 and missingness $< 10\%$. The more permissive MAF and missingness thresholds for SVs reflect their lower abundance and greater genotyping uncertainty inherent to short-read-based SV detection [Ho et al., 2020]. This retained 18,519 variants genome-wide.

Chromosomes were processed in parallel (11 concurrent jobs) and per-chromosome VCFs were merged using `bcftools concat`. Merged VCFs were converted to PLINK2 binary format (`.pgen/.pvar/.psam`) using `plink2` v2.00a [Chang et al., 2015], with the `-new-id-max-allele-len 300 missing` flag to accommodate long allele string representations in INDEL and SV records.

Linkage disequilibrium pruning

To produce sets of approximately independent markers for principal component analysis, we performed LD pruning on SNVs and INDELs using `plink2 -indep-pairwise` with a window of 50 variants, a step size of 5, and a pairwise r^2 threshold of 0.2, following standard practice for population structure analyses [Abdellaoui et al., 2013, Price et al., 2006]. This retained 2,291,979 LD-pruned SNVs and 1,190,449 LD-pruned INDELs. SVs were not subjected to LD pruning because their sparse genomic distribution ($\sim$18,500 variants across 22 autosomes, or roughly one per 165 kb) renders inter-variant LD negligible.

Principal component analysis

PCA was performed independently for each variant type using `plink2 -pca 20`, extracting the top 20 principal components from the genetic relationship matrix. All 3,202 samples were included in each analysis. For SNVs and INDELs, PCA was computed on the LD-pruned marker sets; for SVs, PCA was computed on the full post-filter set of 18,519 variants. The first two PCs captured the majority of genetic variance for all three types (SNV PC1: 54.4%, PC2: 24.7%; INDEL PC1: 54.6%, PC2: 23.3%; SV PC1: 50.7%, PC2: 20.6%).

Procrustes analysis of PCA concordance

To quantify the geometric concordance of population structure across variant types, we applied Procrustes superimposition [Gower, 1975] to the first 10 PCs from each pairwise combination of variant types. We selected 10 PCs as this captures both major continental structure (PCs 1–4) and finer sub-continental differentiation (PCs 5–10); results were qualitatively robust to this choice. Each PC matrix ($3{,}202 \times 10$) was standardized to zero mean and unit variance per component prior to analysis. Procrustes analysis finds the optimal rotation, reflection, and uniform scaling that minimizes the sum of squared distances between two point configurations. We report the Procrustes correlation, defined as $1 - d^2$ where d is the Procrustes distance after optimal superimposition, implemented via `scipy.spatial.procrustes` [Virtanen et al., 2020]. Statistical significance was assessed using a permutation test with 1,000 random permutations of sample labels; the p-value was computed as the proportion of permuted Procrustes correlations exceeding the observed value.

Convergence analysis

To distinguish genuine biological discordance from the statistical effects of unequal variant counts, we developed a convergence analysis framework. LD-pruned SNVs were randomly subsampled (without replacement) to nine target counts spanning three orders of magnitude: 500; 1,000; 5,000; 10,000; 18,519 (matching the SV count); 50,000; 100,000; 500,000; and 1,000,000. At each target count, PCA was recomputed on the subsampled SNV set using `plink2 -pca 20`, and Procrustes concordance was measured against the full INDEL and full SV PCA embeddings. Three independent random seeds (42, 123, 456) were used at each count to assess sampling variability. The rationale is straightforward: if cross-type discordance were driven solely by low variant count, concordance should increase monotonically with the number of variants and eventually converge to a shared ceiling. A plateau in concordance that differs between type pairs—despite increasing variant count—would indicate an irreducible type-specific signal.

Population differentiation (F_{ST})

Pairwise F_{ST} was computed for all 325 unique population pairs ($\binom{26}{2}$) for each variant type using the Hudson estimator [Hudson et al., 1992], which is well-suited for genome-wide ratio-of-averages calculations and performs robustly with unequal sample sizes [Bhatia et al., 2013]:

$$\hat{F}_{\mathrm{ST}} = \frac{(p_1 - p_2)^2 - \frac{p_1(1-p_1)}{n_1-1} - \frac{p_2(1-p_2)}{n_2-1}}{p_1(1-p_2) + p_2(1-p_1)} \tag{1}$$

where p_i and n_i denote the alternate allele frequency and diploid sample size in population i, respectively. Genome-wide F_{ST} was obtained as the ratio of

summed numerators to summed denominators across all variants, which is equivalent to a weighted average that gives more weight to higher-heterozygosity loci. Per-locus estimates were floored at zero. Per-population allele frequencies were computed using `plink2 -freq`. Cross-type concordance of F_{ST} values was quantified using Pearson and Spearman rank correlation coefficients across all 325 population pairs.

Size-stratified F_{ST} analysis

To test whether population differentiation scales with variant size, we partitioned INDELs and SVs by the absolute difference between reference and alternate allele lengths into size bins: 1 bp ($n = 1{,}470{,}230$), 2–5 bp ($n = 1{,}299{,}415$), 6–20 bp ($n = 494{,}238$), and 21–50 bp ($n = 64{,}753$) for INDELs; and 50–200 bp ($n = 17{,}550$) for SVs. The vast majority of SVs in our filtered call set fell within the 50–200 bp range, reflecting the size distribution of short-read-detectable structural variants. For each size bin, we extracted the corresponding variants, computed per-population allele frequencies, and calculated genome-wide F_{ST} across all 325 population pairs as described above.

Structural variant subtype analysis

SVs were classified into deletions (DEL; alternate allele shorter than reference, $n = 14{,}091$) and insertions (INS; alternate allele longer than reference, $n = 4{,}428$) based on allele length comparison. Per-subtype F_{ST} was computed across all 325 population pairs, and per-subtype site frequency spectra were characterized within each super-population, to test whether the direction of the structural rearrangement modulates population differentiation and allele frequency dynamics.

Site frequency spectrum analysis

The site frequency spectrum (SFS) was characterized for each variant type within each of the five super-populations. For each variant, we computed the minor allele frequency (MAF) as $\min(p, 1-p)$ from the super-population-specific allele frequency. We summarized the SFS using mean MAF, the proportion of rare variants (MAF < 0.05), and distributional skewness. Pairwise comparisons of SFS shape between variant types within each super-population were performed using the two-sample Kolmogorov-Smirnov (KS) test, with correction for multiple testing via the Benjamini-Hochberg false discovery rate (FDR) procedure at $\alpha = 0.05$ [Benjamini and Hochberg, 1995]. To control for the different MAF thresholds applied during variant extraction (0.01 for SNVs and INDELs versus 0.005 for SVs), we repeated the SFS analysis applying a uniform MAF ≥ 0.01 filter to all three variant types.

Population tree reconstruction

To visualize population relationships implied by each variant type, we constructed UPGMA (unweighted pair group method with arithmetic mean) dendrograms from the 26×26 pairwise F_{ST} distance matrices using `scipy.cluster.hierarchy` [Virtanen et al., 2020]. Cross-type topological concordance was assessed via the Pearson correlation of the upper triangles of pairwise F_{ST} distance matrices.

Mantel test

To assess the correlation between pairwise genetic distance matrices derived from different variant types, we performed Mantel tests [Mantel, 1967] with permutation-based significance testing. Pairwise Euclidean distance matrices were computed in the space of the first 10 PCs for each variant type. The observed Pearson correlation between distance matrices from two variant types was compared to a null distribution generated by 999 random permutations of sample labels, which accounts for the non-independence inherent in pairwise distance comparisons.

Software and reproducibility

All analyses were performed on a Linux server with 128 CPU cores and 1 TB RAM. Key software versions: `bcftools` 1.17, `plink2` v2.00a, Python 3.10 with `numpy` 1.24, `scipy` 1.11, `pandas` 2.0, `matplotlib` 3.7, and `seaborn` 0.12. Random seeds (42, 123, 456) were used for all stochastic analyses to ensure reproducibility. Complete analysis scripts are available in the project repository.

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Supporting information

S1 Fig. Site frequency spectrum distributions per super-population and variant type. Detailed SFS histograms showing the distribution of minor allele frequencies for SNVs, INDELs, and SVs within each of the five super-populations (AFR, EUR, EAS, SAS, AMR). SVs are dramatically skewed toward rare variants across all populations.

S2 Fig. Count-matched PCA concordance. PCA embeddings when all three variant types are subsampled to 18,519 variants (the SV count), demonstrating that SNV-INDEL concordance ($r \approx 0.65$) exceeds SNV-SV concordance ($r \approx 0.55$) even at matched variant counts.

S3 Fig. SV subtype site frequency spectra. Rare-variant proportion (MAF < 0.05) for deletions and insertions across five super-populations, showing that insertions are consistently more rare across all populations.

S4 Fig. Variant size distributions. Histograms of INDEL sizes (1–50 bp, linear scale) and SV sizes (50–200+ bp, log scale), confirming that the SV call set is dominated by variants in the 50–200 bp range.

S5 Fig. Uniform MAF filter SFS comparison. Site frequency spectra computed with a uniform MAF ≥ 0.01 filter applied to all three variant types, confirming that the SV rare-variant enrichment (59–63% with MAF 0.01–0.05) is biological rather than an artifact of differential MAF thresholds.